

Agentic Data and Knowledge Ecosystem for Healthcare

Carl Yang @ CS, Emory



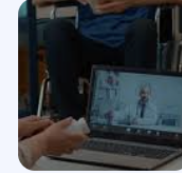
Data Challenges in Healthcare

- Format misalignment
- Errors and other quality issues
- Lack of interoperability
- Limited longitudinal data

Inconsistent Data Formats and Patient Privacy Concerns ...

Oct 30, 2024 — The State of EHR Interoperability. While EHR adoption has...

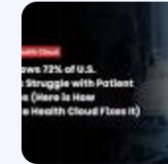
 Simbo AI ⋮



Study Shows 72% of U.S. Hospitals Struggle with Patient Data Gaps ...

Feb 13, 2025 — 1. Lack of Interoperability Among EHR Systems · The health care...

 CapMinds ⋮



Multimodal missing data in healthcare - ScienceDirect.com

* Overview of healthcare data. In healthcare, data acquisition is diverse,...

 ScienceDirect.com ⋮



The Absence of Longitudinal Data Limits the Accuracy of High ...

However, the role of longitudinal data on the accuracy of HTCP algorithms continues to be a concern. Insufficient...

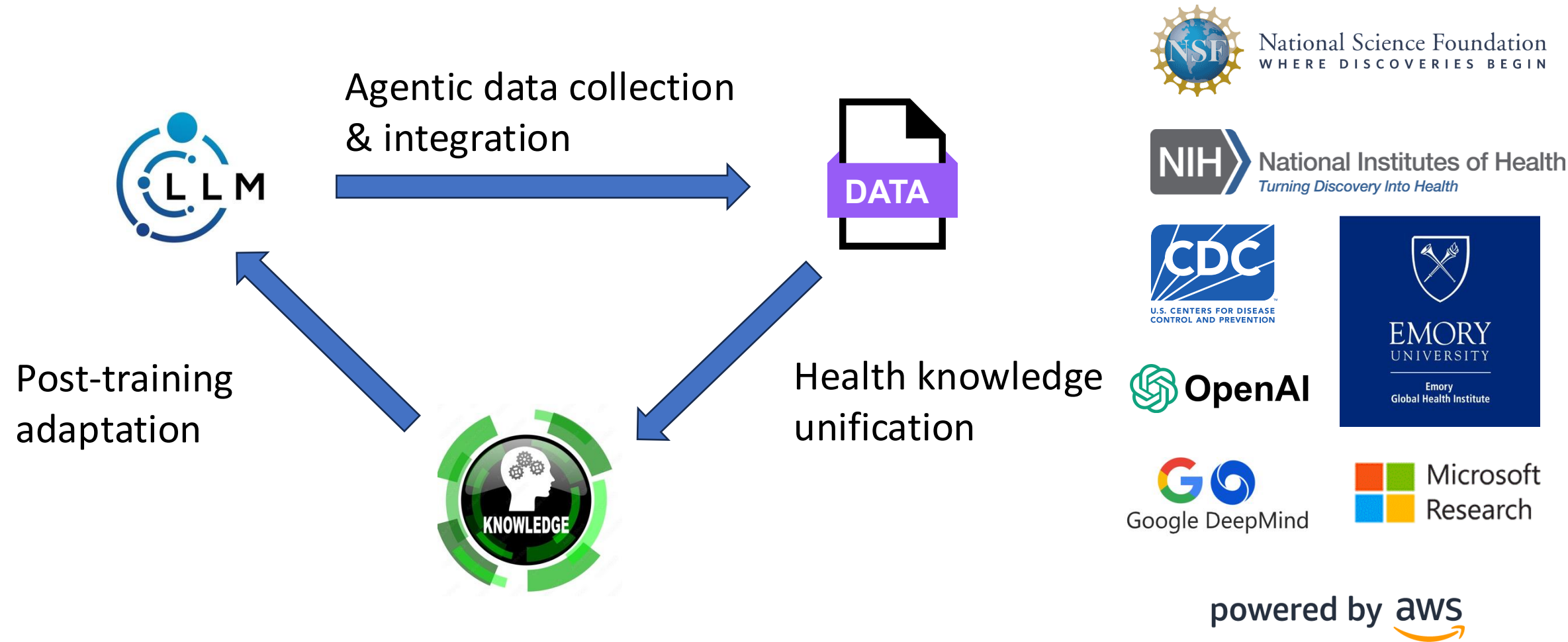
 National Institutes of Health (NIH) | (.gov) ⋮

Model Challenges in Healthcare



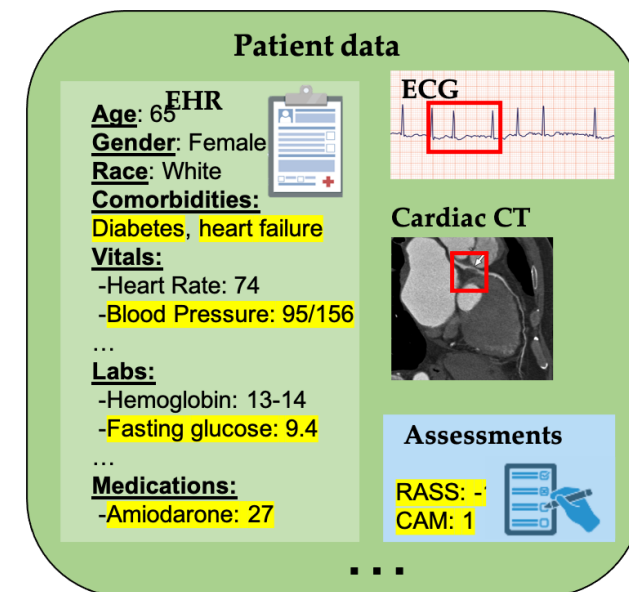
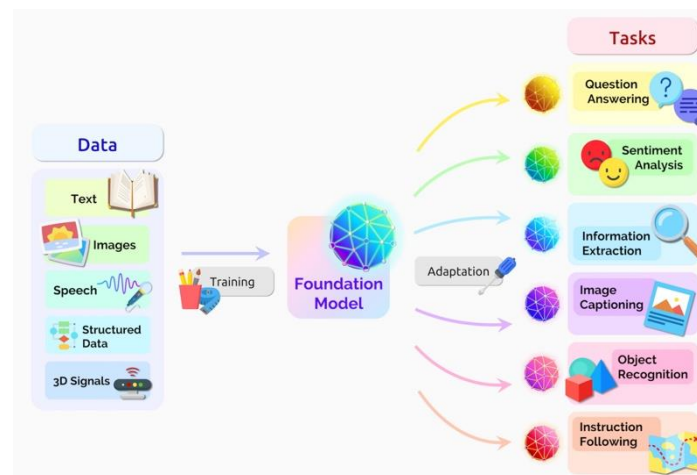
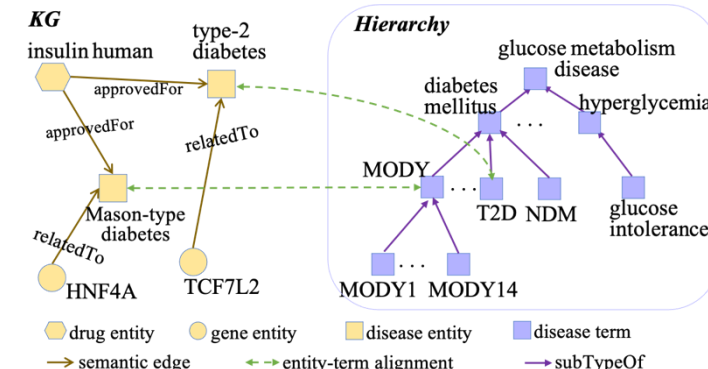
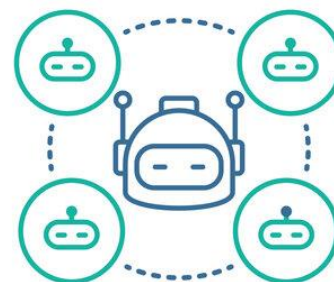
- Medical and clinical knowledge (e.g., subtypes, complications, rare diseases)
- Comprehensive reasoning (e.g., interconnections, multimodalities)
- Reliable validation (Humans? LLMs? Simulations?)
- Precision and privacy trade-off

The Agent-Data-Knowledge Ecosystem



LLM-empowered Data Collection and Integration

- Agentic data pipeline
 - Collection, standardization, linking (KB), generation
- Language as a bridge for multi-modality patient data
 - Reconstruction and resolution
 - Missing data imputation
 - Modality alignment



LLM-empowered Knowledge Discovery

- Unified health knowledge base
 - LLM-facing knowledge base
 - Novel definitions, formulations and standardization process
 - Construction, expansion and consolidation
 - Causal inference [KDD'2025]
 - Rule learning [WWW'2026]
 - Uncertainty
 - Human-in-the-loop

AFib is diagnosed and managed by the cardiology department
Medicaid is suitable for low-income individuals under 65
CABG is a surgical procedure that treats CAD
Patients should watch out for infections after CABG
Emory hospitals offer the world's leading CABG surgery
...

[< Agenda](#)

[Details](#)

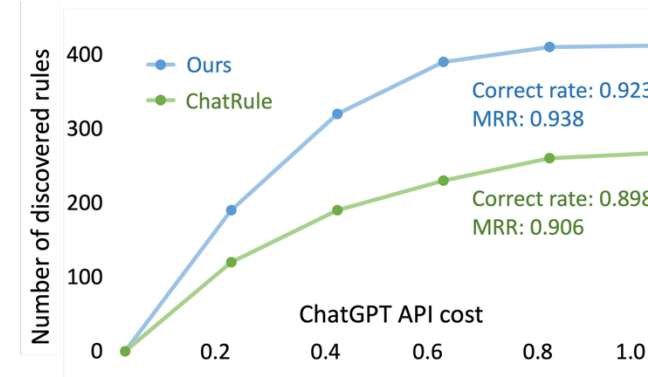


Causal Discovery through Synergizing
Large Language Model and Data-
Driven Reasoning

Wednesday, August 6, 2025
5:10 PM - 5:30 PM

713 A [View map](#)

● Research Track



Knowledge-guided Post-training Techniques

11/19/2025 | 8:00 AM – 9:15 AM | A708

S100: Model Mayhem: Adventures in Interpretability, Reasoning, and the Edge Cases of Care

KERAP: A Knowledge-Enhanced Reasoning Approach for Accurate Zero-shot Diagnosis Prediction Using Multi-agent LLMs

Speaker:

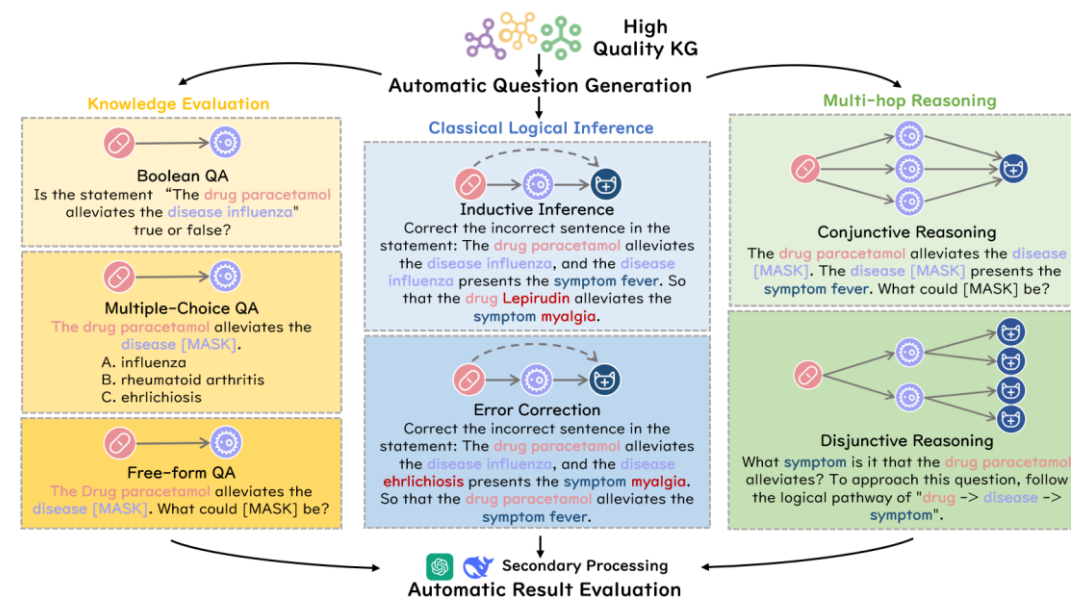
Yuzhang Xie, Doctoral Student

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Authors:

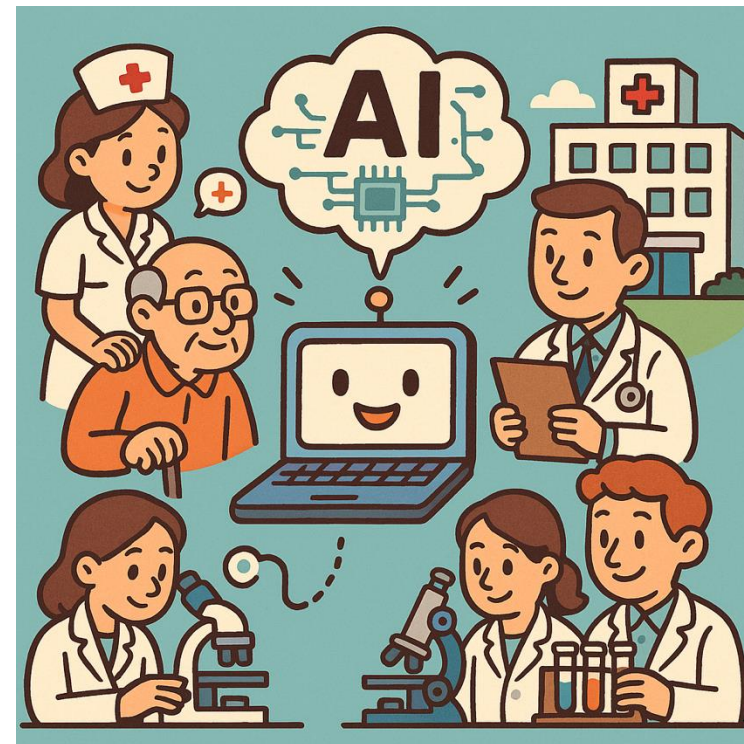
Yuzhang Xie, Doctoral Student - Emory University; Hejie Cui, Postdoc - Stanford University; Ziyang Zhang, Doctoral Student - Emory University; Jiaying Lu, PhD - Emory University School of Nursing's Center for Data Science; Kai Shu, PhD - Emory University; Fadi Nahab, MD - Emory University; Xiao Hu, PhD - Emory University; Carl Yang, PhD;

- Fine-tuning
 - Knowledge-generated instruction [AMIA'2025]
 - Structure-guided reward model [ICLR'2025]
- Inference-time reasoning
 - Knowledge-infused prompt (RAG) [KDD'2025]
 - Structure-enhanced self-inference [AMIA'2024]
- Validation
 - Automated benchmarking [AI Open'2025]
 - Post-hoc error detection [npj Digital Medicine'2024]



Clinical Applications and Deployments

- Holistic and personalized clinical workflow optimization
 - Comprehensive diagnosis and care (multimodality, progressive)
 - Personalized complication analysis (clustering and subtyping)
 - Multi-task holistic CDS (risk prediction, treatment suggestion, outcome analysis and longitudinal modeling)
- Who will benefit
 - Patients: personalized diagnosis reference and risk/outcome management
 - Doctors: diagnostic and following-up assistance
 - Hospitals: standardized clinical workflows and advert event control
 - Medical centers: AI-driven biomedicine and healthcare research



Ack.



Thank you!

- Questions & discussions
- Welcome to collaborate!
 - j.carlyang@emory.edu
 - <https://cs.emory.edu/~jyang71/>
- An ad for KDD AI4Sciences
 - <https://kdd2026.kdd.org/ai4sciences-track-call-for-papers/>

